01 / Neural control and motility

Lecture PDF pp. 4–10 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
A1 Local ENS input Distension / chemicals in lumen	GI mucosal mechanoreceptors / chemoreceptors → local ENS circuits	MYENTERIC plexus → motility SUBMUCOSAL plexus → secretion	Outputs reach smooth muscle, secretory cells and endocrine cells.	Local coordination. ENS can function independently; autonomic input modulates it.
A2 Parasympathetic input	Vagal pathways → enteric neurons	ACh and other enteric outputs	Generally stimulates GI smooth muscle and secretory activity.	Motility ↑; secretion ↑. Responses differ among specific muscles / sphincters.
A3 Sympathetic input	Sympathetic pathways acting on the GI / ENS network	Sympathetic modulation of enteric and effector activity	Generally suppresses GI activity.	Motility ↓; secretion ↓.
A4 Pacemaker activity	INTERSTITIAL CELLS OF CAJAL (ICCs), myenteric region	Spontaneous slow waves GAP JUNCTIONS couple ICCs to smooth muscle	Slow waves spread → muscle depolarisation. L-type Ca^{2+} channels permit contraction†.	Coordinates contraction timing. A slow wave does not necessarily produce a strong contraction†.
A5 Food enters stomach	Orad stomach: fundus / proximal body Caudad stomach: distal body / antrum	Orad relaxation; distal contractions; pyloric delivery	Receive bolus → grind / mix with gastric juice → propel chyme through pylorus.	Accommodation → mixing / digestion → gastric emptying into duodenum.
A6 Food in small intestine	Circular-muscle segments with neighbouring relaxation	SEGMENTATION	Alternating contraction → split chyme → move backward / forward → merge again.	Primarily MIXING; improves enzyme / mucosal contact rather than sustained net propulsion.
A7 Muscle contraction pattern	MUSCULARIS EXTERNA Circular + longitudinal layers	Circular contraction / longitudinal contraction Phasic / tonic patterns	Circular → diameter ↓; longitudinal → segment length ↓.	Phasic = repeated contraction / relaxation, e.g. mixing. Tonic = maintained tone, e.g. sphincters.

Tutorial links: MCQ 1–2; Case 1 Q2–4; Case 2 Part 1 Q3.

Structure check: Myenteric plexus lies BETWEEN circular and longitudinal muscle. Submucosal plexus lies in the submucosal region, internal to circular muscle.

02 / Start signals and the three acid stimulators

Lecture PDF pp. 11–14, 16 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
B1 Sight / smell / taste / anticipation of food	CNS → parasympathetic VAGAL output → gastric enteric neurons	ACh → gastric effectors GRP → G cells (B6)	Prepares gastric secretion and motility BEFORE a meal enters the stomach.	CEPHALIC PHASE: feedforward control. Gastric acid / motility ↑; reduces delay after eating.
B2 Gastric distension	Gastric mechanoreceptors → long VAGOVAGAL reflex	Vagal / enteric output to stomach	Promotes gastric secretion and motor activity.	GASTRIC PHASE: major phase in the lecture's comparison.
B3 Gastric distension	Local gastric ENS reflex	Enteric ACh release ↑	Acts locally on smooth muscle and gastric secretory pathways.	Gastric motility ↑; secretion ↑.
B4 Vagal / enteric activation	Gastric ENTERIC NEURONS	ACh → M_3 receptor on PARIETAL cell	APICAL H^+/K^+-ATPase activity ↑ → H^+ secreted into gastric lumen.	HCl secretion ↑. Neurocrine route.
B5 ECL activation	ECL cells in gastric mucosa → nearby parietal cells	HISTAMINE → parietal H_2 receptor	APICAL H^+/K^+-ATPase activity ↑ → H^+ secretion ↑.	HCl secretion ↑. Paracrine route.
B6 Vagal / enteric GRP input	GRP stimulates antral G CELLS → gastrin enters blood	GASTRIN → parietal CCK_2 receptor (CCK-B is another name)	Circulating gastrin stimulates the parietal proton-pump pathway.	HCl secretion ↑. Endocrine route.
B7 ACh OR gastrin	Enteric ACh; circulating gastrin from G cells	ECL stimulation Gastrin acts via CCK_2 on ECL cells	Histamine release ↑ → H_2 on nearby parietal cells → proton pump ↑.	INDIRECT acid stimulation. ACh / gastrin also have direct parietal routes (B4 / B6).

Tutorial links: Case 1 Q1–6; Case 2 Part 1 Q1–3 and Part 2 Q1 / Q3.

Important source correction: Slide 12 and a handwritten tutorial answer list low gastric pH as a stimulatory gastric-phase input. Strong gastric acidity instead activates D-cell somatostatin and inhibits acid secretion (C1–C2). Gastric distension supports stimulation; meal peptides / amino acids also promote G-cell gastrin release†.

Keep the two neural outputs distinct: ACh stimulates parietal / ECL pathways; GRP is the expected neural messenger when specifically asking how vagal input stimulates G-cell gastrin release.

03 / Gastric acid inhibition and drug targets

Lecture PDF pp. 15–16 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
C1 Gastric acidity ↑ Gastric pH ↓	Gastric D CELLS	Somatostatin → nearby PARIETAL cells	Direct inhibition of acid-secretory activity / proton-pump drive.	HCl secretion ↓. Paracrine inhibitory route.
C2 Gastric acidity ↑ Gastric pH ↓	Gastric D CELLS	Somatostatin → G CELLS and ECL CELLS	Gastrin release ↓ AND histamine release ↓ → parietal stimulation ↓.	HCl secretion ↓ through INDIRECT routes.
C3 Local prostaglandins	Gastric mucosal signalling (source summary diagram)	Prostaglandins → inhibitory receptor on parietal cell	Acid-secretory / proton-pump drive decreases.	H^+ secretion ↓. No new receptor subtype is required by this lecture diagram.
C4 Atropine	Muscarinic receptor pathway	Muscarinic blockade, including parietal M_3	Reduces ACh-mediated stimulation of acid secretion.	HCl response to ACh ↓. Histamine / gastrin receptors are separate targets.
C5 Cimetidine	PARIETAL cell	H_2 receptor blockade	Histamine-mediated parietal stimulation ↓.	Acid secretion ↓. M_3 / CCK_2 inputs can still signal.
C6 Omeprazole	APICAL membrane of PARIETAL cell	H^+/K^+-ATPase inhibition	Blocks the final acid-secretory pump despite upstream receptor signals.	H^+ secretion ↓. Final common target of the stimulatory routes.

Tutorial links: MCQ 3; Case 2 Part 2 Q2.

NEGATIVE FEEDBACK: HCl ↑ → gastric pH ↓ → D cells / somatostatin ↑ → parietal stimulation and HCl secretion ↓. Reduced acidity then reduces the original D-cell stimulus.

Meal buffering: Food initially buffers H^+. As food leaves the stomach, buffering falls and pH drops more → somatostatin inhibition becomes stronger.

Interpretation: ACh, gastrin and histamine cooperate to stimulate acid; cooperation alone is not positive feedback. Cephalic preparation is feedforward, while the low-pH D-cell loop is negative feedback.

04 / Duodenal control of gastric emptying

Lecture PDF p. 17 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
D1 Lipids, acid and distension IN DUODENUM	Duodenal sensing → ENTEROGASTRIC reflex	Neural inhibition: sympathetic suppression / reduced vagal gastric stimulation in source figure	Gastric motility ↓ AND pyloric-sphincter contraction ↑.	Gastric emptying slows → duodenum receives a smaller incoming load.
D2 Amino acids, peptides, fatty acids	I CELLS in DUODENUM + JEJUNUM	CCK → blood → gastric motor / neural targets	Inhibits gastric motility / delivery.	Gastric emptying ↓. More time for small-intestinal digestion / absorption.
D3 Low-pH chyme IN DUODENUM	S CELLS in duodenal epithelium	SECRETIN → blood → gastric motor targets	Inhibits gastric motor activity / delivery in the lecture model.	Gastric emptying ↓. Incoming acid delivery is limited.
D4 Early intestinal phase	Intestinal endocrine input (source figure caption)	Intestinal GASTRIN → blood → stomach	Brief gastric stimulation, followed by predominant CCK / secretin / enterogastric inhibition.	Transient early stimulation; later intestinal inhibition is the dominant teaching emphasis.

Tutorial links: MCQ 6; Case 2 Part 3 Q1–2 / Q5.

NEGATIVE FEEDBACK: The duodenal brake limits upstream gastric delivery while the duodenum neutralises acid and processes chyme. As that load resolves, inhibitory drive can ease.

Location changes the answer: GASTRIC distension generally stimulates gastric-phase activity; DUODENAL distension contributes to inhibition of gastric emptying.

Parallel hormone effects: CCK also stimulates pancreatic enzymes (E1). Secretin also stimulates bicarbonate-rich fluid (E3). The gastric and pancreatic effects are parallel outputs, not a sequence in which one hormone produces the other.

05 / Pancreatic enzyme versus aqueous secretion

Lecture PDF pp. 18–19 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
E1 Amino acids, peptides, fatty acids	I CELLS: DUODENUM + JEJUNUM → CCK into blood	CCK → CCK_1-mediated pathway of EXOCRINE PANCREAS	Pancreatic ACINAR-cell enzyme-rich secretion ↑.	More pancreatic digestive enzymes delivered to intestinal lumen.
E2 Neural stimulation	Vagal / enteric cholinergic input to pancreas	ACh → pancreatic muscarinic pathway	Promotes acinar enzyme secretion; enhances CCK-associated secretion.	Enzyme output ↑. ACh is an additional input, not the main enzyme-stimulating hormone.
E3 Duodenal pH ↓	S CELLS in DUODENAL EPITHELIUM → secretin into blood	Secretin → pancreatic DUCTAL-cell receptor	HCO_3^--rich aqueous secretion ↑; Na^+ / water accompany the fluid.	Gastric acid neutralised in duodenum → pH rises toward a suitable digestive range.
E4 Cholinergic and CCK input	ACh / circulating CCK acting with secretin	ACh + CCK potentiate pancreatic aqueous secretion	Increase / support ductal HCO_3^--rich fluid output in the source diagram.	Secretin remains the PRIMARY hormone for aqueous output in the tutorial comparison.

Tutorial links: MCQ 4–5; Case 2 Part 3 Q3–4.

NEGATIVE FEEDBACK: Duodenal pH ↓ → S cells → secretin ↑ → ductal HCO_3^- secretion ↑ → acid neutralisation → the low-pH stimulus decreases → secretin drive falls.

Three distinctions: CCK_1 = pancreatic enzyme pathway; CCK_2 = gastrin-associated gastric acid pathway. ACINAR cells supply enzymes; DUCTAL cells supply bicarbonate-rich fluid. Gastric D cells and duodenal S cells respond to acidity with different outputs.

Figure details retained: The CCK figure gives phenylalanine, methionine, tryptophan, small peptides and fatty acids as examples. Second messengers are deliberately omitted here.

06 / Carbohydrate digestion and lactose intolerance

Lecture PDF pp. 20–21 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
F1 Starch	Saliva (minor role) and PANCREATIC JUICE (major luminal role)	α-AMYLASE	Luminal starch digestion → α-dextrins, maltose and maltotriose.	Starch fragments still require brush-border digestion before absorption.
F2 α-Dextrins	SMALL-INTESTINAL BRUSH BORDER	α-DEXTRINASE / isomaltase	Final hydrolysis of starch-derived fragments.	Glucose produced.
F3 Maltose / maltotriose	SMALL-INTESTINAL BRUSH BORDER	MALTASE; overlapping brush-border α-glucosidases Source maltotriose arrow labels sucrase	Final hydrolysis to monosaccharides.	Glucose produced.
F4 Sucrose	SMALL-INTESTINAL BRUSH BORDER	SUCRASE	Sucrose hydrolysed.	Glucose + fructose.
F5 Lactose	SMALL-INTESTINAL BRUSH BORDER	LACTASE	Lactose hydrolysed.	Glucose + galactose.
F6 Trehalose	SMALL-INTESTINAL BRUSH BORDER	TREHALASE	Trehalose hydrolysed.	Two glucose units.
F7 Lactase deficiency + lactose intake	Lactose remains in intestinal lumen	Inadequate lactase → lactose undigested / unabsorbed	Luminal solute retains water. Bacterial fermentation may add gas / bloating†.	OSMOTIC DIARRHOEA. This is a digestion defect upstream of monosaccharide transport.

Tutorial links: Case 2 Part 4 Q1; apply enzyme-versus-transporter distinctions.

Important source correction: The source table / text attribute α-amylase to intestinal mucosa. Use salivary / pancreatic α-amylase for LUMINAL starch hydrolysis; BRUSH-BORDER disaccharidases and α-dextrinase finish digestion. Only monosaccharides are absorbed in this pathway.

Process type: Hydrolysis and lactose-induced osmotic water retention are process sequences, not automatically feedback loops.

07 / Monosaccharide absorption: membrane locations

Lecture PDF p. 22 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
G1 Energy for Na^+-coupled uptake	BASOLATERAL membrane of intestinal ENTEROCYTE	Na^+/K^+-ATPase uses ATP	Na^+ moves out; K^+ moves in → low intracellular Na^+ maintains the inward Na^+ gradient.	Na^+ gradient provides energy for APICAL SGLT1. SGLT1 itself does not hydrolyse ATP.
G2 Glucose OR galactose IN LUMEN	APICAL / brush-border membrane of enterocyte	SGLT1: Na^+–sugar SYMPORT	Na^+ and glucose / galactose enter together; sugar can move against its gradient.	Glucose / galactose enters enterocyte by SECONDARY ACTIVE transport.
G3 Fructose IN LUMEN	APICAL / brush-border membrane of enterocyte	GLUT5	Na^+-independent facilitated diffusion into cell.	Fructose enters enterocyte. Different apical route from glucose / galactose.
G4 Glucose, galactose and fructose INSIDE CELL	BASOLATERAL enterocyte membrane	GLUT2	All three monosaccharides leave toward interstitial fluid / intestinal capillaries.	Intestinal blood → portal circulation†. GLUT2 is the EXIT route in the lecture model.

Tutorial links: Case 2 Part 4 Q2: “from the lumen” requires APICAL SGLT1.

One complete glucose route: Lumen → APICAL SGLT1 with Na^+ → enterocyte → BASOLATERAL GLUT2 → blood. BASOLATERAL Na^+/K^+-ATPase maintains the gradient that powers the entry step.

Source comparison: The slide calls Na^+-coupled glucose / galactose uptake “higher efficiency” and fructose uptake “relatively inefficient”. The essential distinction is coupled uphill transport versus facilitated diffusion, not a universal numerical ranking.

Process type: ATP-to-Na^+-gradient coupling supplies energy. It is not positive feedback. Apical faces lumen; basolateral faces interstitium / blood.

08 / Protein digestion and other gastric products

Lecture PDF pp. 11, 20 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
H1 Gastric protein / pepsinogen	BODY chief cells secrete pepsinogen; BODY parietal cells supply HCl	Acid supports conversion of PEPSINOGEN → PEPSIN	Pepsin begins protein hydrolysis in gastric lumen; acid also denatures proteins.	Protein digestion initiated. HCl is acid; pepsin is the enzyme.
H2 Pancreatic protease activation†	Duodenal BRUSH BORDER; pancreatic trypsinogen in lumen	ENTEROKINASE = ENTEROPEPTIDASE	Activates trypsinogen → trypsin. Trypsin activates further pancreatic protease precursors†.	Active pancreatic protease system. Enteropeptidase activates a zymogen, rather than acting as the main bulk protease.
H3 Luminal protein / peptides	PANCREATIC secretion into small intestine	TRYPSIN; CHYMOTRYPSIN; CARBOXYPEPTIDASE	Hydrolyse proteins / peptides into smaller peptides and amino-acid products.	Products prepared for further mucosal digestion / absorption.
H4 Small peptides	INTESTINAL MUCOSA / brush-border processing	AMINO-OLIGOPEPTIDASE; DIPEPTIDASE	Further peptide hydrolysis. Intracellular peptidases also contribute†.	Smaller peptide / amino-acid products. Detailed protein transporters are outside this lecture's absorption focus.
H5 Vitamin B_{12}	PARIETAL cells → INTRINSIC FACTOR into gastric lumen	Intrinsic factor associates with vitamin B_{12}†	Complex travels to ILEUM, where intrinsic factor facilitates B_{12} uptake.	Vitamin B_{12} absorption. Same parietal cell also secretes HCl, but the products have different functions.
H6 Gastric surface protection	Gastric MUCOUS cells → mucus into lumen / surface layer	MUCUS	Protective, lubricating surface layer limits acid injury.	Mucosal protection + lubrication. Not a digestive enzyme or endocrine hormone.

Tutorial links: Case 2 Part 2 Q3–4: parietal HCl and its digestive role.

Cell-location clarification: The source table lists parietal / chief cells in the body and G / mucous cells in the antrum. Mucus-producing cells are not confined to the antrum throughout the stomach. Gastrin goes to BLOOD; HCl, pepsinogen, intrinsic factor and mucus go to the LUMEN.

Source versus explanation: The enzyme names are retained from slide 20. † flags explanatory activation / processing steps where the source lists the enzyme but does not spell out the complete mechanism.

09 / Lipid digestion: bile and pancreatic enzymes

Lecture PDF pp. 20, 23 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
I1 Large dietary lipid droplets	LIVER makes bile; GALLBLADDER stores / releases it → duodenal lumen	BILE SALTS: amphipathic, with hydrophobic and hydrophilic faces	EMULSIFICATION into fine droplets → enzyme-contact surface area ↑.	Physical dispersion aids digestion. Bile salts are not hydrolytic enzymes.
I2 Triglyceride (TG)	PANCREATIC secretion acting in small-intestinal lumen	PANCREATIC LIPASE + COLIPASE	Lipase hydrolyses TG; colipase assists action at the lipid–water interface.	Monoglyceride + two FREE FATTY ACIDS (FFAs).
I3 Cholesterol ester	PANCREATIC secretion → intestinal lumen	CHOLESTEROL ESTER HYDROLASE	Hydrolyses cholesterol ester.	Cholesterol + FFA.
I4 Phospholipid	PANCREATIC secretion → intestinal lumen	PHOSPHOLIPASE A_2	Hydrolyses phospholipid.	Lysolecithin / lysophospholipid + FFA.
I5 Preliminary oral lipid enzyme	LINGUAL secretion (listed in source enzyme table)	LINGUAL LIPASE	Contributes to preliminary triglyceride digestion†.	The main small-intestinal lipid pathways in this lecture are I1–I4.

Tutorial links: Case 2 Part 4 Q3: FFAs / monoglycerides are the main expected pair.

Two separate operations: EMULSIFICATION disperses droplets and increases surface area. HYDROLYSIS breaks lipid chemical bonds using enzymes. Neither is itself a negative-feedback loop.

Product checklist: The products entering mixed micelles include FFA, monoglycerides, cholesterol and lysolecithin / lysophospholipids. These are not yet chylomicrons.

10 / Lipid absorption: lumen → cell → lymph → blood

Lecture PDF pp. 24–25 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

STIMULUS / MATERIAL →	SOURCE / LOCATION →	MESSENGER / RECEPTOR / PROTEIN →	TARGET / ACTION →	EFFECT / INTERPRETATION
J1 Lipid digestion products IN LUMEN	Small-intestinal lumen → brush border	BILE-SALT MIXED MICELLES carry FFA, MG, cholesterol and LysoPL	Lipid cargo leaves micelle at APICAL surface and enters enterocyte. Whole micelles do NOT enter.	Lipid products enter the cell. Bile salts remain in lumen for subsequent ileal absorption.
J2 Lipid products INSIDE ENTEROCYTE	Enterocyte cytoplasm	RE-ESTERIFICATION with fatty-acid chains	MG → TG; cholesterol → cholesterol ester; LysoPL → phospholipids.	Complex lipids rebuilt inside cell. This differs from luminal digestion.
J3 Re-esterified lipids	INSIDE ENTEROCYTE	APOPROTEINS, including APO B	TG / cholesterol esters / phospholipids packaged with apoproteins.	CHYLOMICRONS formed. Carrier differs from a luminal micelle.
J4 Chylomicrons	BASOLATERAL exocytosis from enterocyte	LACTEALS → lymphatic circulation → THORACIC DUCT	Chylomicrons travel through lymph and enter systemic blood.	Main long-chain dietary lipid route to the circulation.
J5 Bile salts left in intestinal lumen	ILEUM; predominantly terminal ileum†	Bile-salt reabsorption	Portal blood → liver → secretion into bile for reuse†.	ENTEROHEPATIC RECYCLING†. Recycling is not automatically positive feedback.

Tutorial links: MCQ 7; Case 2 Part 4 Q4: carrier = chylomicron; anatomical route = lacteals / lymph / thoracic duct.

Scope of the route: This is the major LONG-CHAIN dietary lipid pathway. Short- / medium-chain fatty acids can reach portal blood more directly†. The source's schematic "MG + FFA → TG" represents addition of further fatty-acid chains, not a one-fatty-acid stoichiometric equation.

Source correction retained: The liver MAKES bile; the gallbladder stores / releases it. Micelles deliver luminal lipid products; chylomicrons export rebuilt lipid from enterocytes.

11 / Cell–messenger–receptor atlas and exam checks

Lecture PDF pp. 5–6, 11, 13–19, 22–25 | Dr Enoch Chan | Read LEFT → RIGHT. † = supporting physiology beyond explicit slide text.

MESSENGER	SOURCE / ROUTE	RECEPTOR / TARGET	MAIN EFFECT
ACh	Enteric neurons; local neurocrine route	Gastric parietal M_3; ECL / muscle / pancreatic cholinergic targets	Promotes acid, motility and pancreatic secretion, depending on target.
GRP	Enteric neurons → gastric G cells	G cell	Gastrin secretion ↑. GRP is distinct from the ACh → parietal route.
Gastrin	G cells, gastric antrum → blood	CCK_2 on parietal / ECL cells	Direct acid stimulation + indirect histamine release.
Histamine	ECL cells in gastric mucosa → local diffusion	H_2 on nearby parietal cells	Acid secretion ↑.
Somatostatin	Gastric D cells → local paracrine actions	Parietal cells; G cells; ECL cells	Direct acid inhibition + gastrin / histamine inhibition.
CCK	I cells, DUODENUM + JEJUNUM → blood	CCK_1 enzyme-secretory pathway; gastric motor / neural targets	Pancreatic enzymes ↑; gastric emptying ↓; potentiates aqueous secretion.
Secretin	S cells, DUODENAL EPITHELIUM → blood	Pancreatic DUCTAL cells; gastric motor targets	HCO_3^--rich fluid ↑; gastric emptying ↓.

Most useful pairs: Myenteric = motility / submucosal = secretion; CCK_1 = pancreatic enzyme route / CCK_2 = gastric gastrin route; acinar = enzymes / ductal = HCO_3^- fluid; SGLT1 = glucose entry / GLUT2 = sugar exit; micelle = lumen / chylomicron = enterocyte export.

Question-style check: For any pathway, name the STIMULUS and LOCATION first, then the source cell, messenger, target receptor / protein and final effect. “Low pH” alone is incomplete: gastric acidity activates D-cell somatostatin; duodenal acidity activates S-cell secretin.

Coverage / source: All seven tutorial MCQs and all 22 case prompts are linked on the relevant pages. Lecture key pathway coverage: pp. 4–10 → A; 11–16 → B/C/H; 17 → D; 18–19 → E; 20–22 → F/G/H/I; 23–25 → I/J; summary p. 26 → atlas. Cover / references add no control pathway.

Scope: This is a pathway companion to the expanded lecture notes, not a replacement for the illustrated source slides. Important names / sites / transporters are retained; tiny intracellular signalling details are omitted. † identifies supporting physiology. Sources: Dr Enoch Chan, BBMS2006 Digestive Physiology and GI Physiology Tutorial 2026–27.